AI-POWERED E-LEARNING PLATFORM

PROJECT REPORT

Submitted by

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Register No: 24MCA019

In the partial fulfilment for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

in

DEPARTMENT OF COMPUTER APPLICATIONS

KUMARAGURU COLLEGE OF TECHNOLOGY

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APRIL 2026

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PROJECT WORK

APRIL 2026

This is to certify that the project entitled

AI-POWERED E-LEARNING PLATFORM

is the bonafide record of project work done by

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Internal Examiner	**External Examiner**

DECLARATION

I affirm that the project work titled **AI-Powered E-Learning Platform** being submitted in partial fulfillment for the award of **Master of Computer Applications** is the original work carried out by me. It has not formed the part of any other project work submitted for award of any degree or diploma, either in this or any other University.

Signature of the Candidate

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I certify that the declaration made above by the candidate is true

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ACKNOWLEDGEMENT

I thank almighty who has been with me through every walk of life, guarding me and showering the blessings throughout the endeavour to put forth my dissertation.

I sincerely thank **SHRI. SHANKAR VANAVARAYAR**, President, Kumaraguru Institutions, for providing the ample facilities to carry out this project.

I sincerely thank **Dr. M. EZHILARASI**, Principal, Kumaraguru College of Technology, for giving me an opportunity to undertake this full project work.

I am grateful to **Dr. V. VIJILESH**, Associate Professor and Head, Department of Computer Applications, Kumaraguru College of Technology for his continuous encouragement and support during the course of this project.

I sincerely thank my guide and Project Coordinator, **Dr. N. JAYAKANTHAN**, Assistant Professor – III, Department of Computer Applications, Kumaraguru College of Technology, for your sincere advice, thought provoking discussions and immense help throughout the project and encouragement given by them.

I am grateful to all my **Senior Managers, Mentors and Colleagues at Codec Technologies** for providing me an opportunity to work in this project and for guiding me and encouraging me always.

I wish to thank my parents, friends and well-wishers who have been a catalyst throughout the development of the project work. Finally, I would like to mention here that I am greatly indebted to each and everybody who has been associated with my project at any stage, but whose name does not find a place in this acknowledgment.

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ABSTRACT

The project titled **"AI-Powered E-Learning Platform"** is a comprehensive web application developed to transform traditional online education through the integration of artificial intelligence technologies. The platform offers personalized learning experiences by analyzing individual student data. Using machine learning algorithms, the system dynamically recommends courses, study materials, and practice exercises, ensuring that each learner receives content suited to their needs and skill level. Key features of the platform include adaptive assessments, real-time feedback, automated grading, and interactive dashboards for tracking progress. The web application supports multimedia content delivery, including study materials, quizzes, and discussion forums, fostering an interactive and collaborative learning environment.

Additionally, the platform incorporates natural language processing for intelligent tutoring and chatbot support, enabling students to clarify doubts instantly. Designed with scalability in mind, the platform can be deployed across various educational institutions and is accessible via desktop and mobile devices. This project demonstrates the potential of AI to create smarter, more effective, and inclusive online learning solutions for the modern digital era.

Technologies

Frontend: React.js

Backend: Node.js

Database: MongoDB

IDE: Visual studio

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
API	Application Programming Interface
JSX	Java Script XML
JSON	JavaScript Object Notation
CRUD	Create, Read, Update, Delete
DOM	Document Object Model
NPM	Node Package Manager
ENV	Environment (as in .env file)
UI/UX	User Interface/User Experience
ODM	Object Data Modeling
MVP	Minimum Viable Product
JWT	JSON Web Token
REST	Representational State Transfer

CHAPTER 1

INTRODUCTION

1.1 ORGANIZATION PROFILE

EroSkillUp is a prominent educational and technical training institute located in Erode, dedicated to bridging the gap between academic knowledge and industry requirements. Established in response to the rapid evolution of technology and the growing demand for skilled professionals, the institute focuses on equipping students, graduates, and working professionals with relevant and up-to-date technical skills. With a strong emphasis on practical, hands-on learning, EroSkillUp aims to develop competent and job-ready individuals who can succeed in the competitive global IT landscape.

The institute has built a solid reputation for offering high-quality training across a diverse range of modern technology domains. Its programs cover areas such as Full Stack Development, including technologies like React JS, Node.js, PHP, Java, and Spring Boot, along with UI/UX Web Design, Android and iOS App Development, Data Analytics, Python, DevOps, Software Testing, and Digital Marketing. To further enhance global career prospects, EroSkillUp also provides language training programs, including Japanese, enabling learners to prepare for international opportunities and work environments.

Beyond its core training offerings, EroSkillUp is committed to comprehensive career development by delivering flexible and industry-oriented services. These include real-time internship opportunities, structured placement training, personalized one-on-one mentoring, online learning options, and customized corporate training programs designed for workforce upskilling. The development of the AI-Powered E-Learning Platform aligns with the institute's forward-thinking digital vision, aiming to extend its reach and provide a more accessible, intelligent, and enriched learning experience for its expanding community of learners.

1.2 OBJECTIVE OF THE PROJECT

The primary objective of the AI-Powered E-Learning Platform is to enhance traditional online education by integrating advanced artificial intelligence technologies into the learning process. This project aims to move beyond the conventional "one-size-fits-all" approach by developing a dynamic and adaptive learning environment that caters to the individual needs, learning pace, and preferences of each student. By utilizing machine learning techniques and modern web technologies, the platform seeks to provide a more effective, inclusive, and engaging educational experience.

Another key objective of the project is to automate and improve conventional teaching and administrative processes. This is achieved through the implementation of intelligent features such as AI-based test generation, which automatically creates quizzes from uploaded study materials like PDFs, thereby reducing the manual workload of educators. Additionally, the platform enhances accessibility to multimedia content by incorporating natural language processing techniques, including automated transcription of video lectures. This feature supports diverse learning styles and improves the searchability of educational content.

The project also focuses on creating an interactive and data-driven learning ecosystem by providing comprehensive tools for both learners and instructors.

- **For Learners:** The platform offers user-friendly dashboards for real-time progress tracking, adaptive assessments tailored to individual performance, and automated evaluation systems that provide instant feedback for continuous improvement.
- **For Instructors/Institutions:** The system provides a centralized and scalable platform for efficient delivery of multimedia content, effective assignment management, and detailed analytics to monitor student engagement and performance.

Overall, the objective is to develop a robust and scalable full-stack web application using technologies such as React.js, Node.js, and MongoDB. The platform demonstrates the practical application of artificial intelligence in building intelligent, accessible, and user-centric digital learning solutions suited for modern educational needs.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

In the current educational landscape, traditional e-learning platforms and Learning Management Systems (LMS) primarily operate as static repositories of educational content rather than interactive learning environments. In these systems, instructors upload study materials such as PDFs, presentations, and pre-recorded video lectures, while students are expected to independently access and understand the content. The learning process typically follows a fixed and linear structure, treating all learners uniformly without considering differences in learning pace, comprehension levels, or individual learning preferences.

Moreover, both administrative and instructional processes in existing systems are largely manual and time-intensive. Instructors are required to spend considerable time creating assessments, preparing quizzes from course materials, and evaluating student submissions. This reduces the time available for effective teaching and student interaction. Additionally, multimedia content, such as video lectures, often lacks interactive features, automated transcription, and efficient search functionality unless these elements are manually implemented. As a result, these systems lack the intelligence required to adapt to individual learner needs and rely heavily on instructors to maintain engagement and accessibility.

2.1.1 Drawbacks of the Existing System

Despite enabling widespread access to educational content, traditional e-learning platforms exhibit several limitations that negatively impact the overall teaching and learning experience:

- **Lack of Personalization:** Existing systems do not support adaptive learning. All students receive identical content and follow the same pace, regardless of their individual understanding or performance levels.
- **High Manual Effort for Educators:** Instructors are burdened with repetitive tasks such as manual quiz creation, assignment evaluation, and progress tracking, which increases workload and reduces efficiency.

- **Limited Accessibility and Searchability:** Video-based content generally lacks automated transcription, making it less accessible for students with hearing impairments or those who prefer text-based learning. Additionally, the absence of searchable transcripts makes it difficult to locate specific information within videos.
- **Delayed Feedback:** The lack of automated grading and intelligent feedback mechanisms results in delays in providing feedback to students, which can hinder continuous learning and timely improvement.

2.2 PROPOSED SYSTEM

The proposed system is an advanced AI-Powered E-Learning Platform designed to address the limitations of traditional Learning Management Systems by incorporating modern artificial intelligence techniques and real-time processing capabilities. Unlike conventional systems that function as static content repositories, this platform provides a dynamic and interactive learning environment tailored to the needs of three primary user groups: learners, instructors, and administrators. The system is developed using a modern technology stack, including a responsive React.js frontend, a robust backend built with Node.js and Express.js, and a scalable MongoDB database. This architecture ensures high performance, efficient media delivery, and a seamless user experience across both desktop and mobile devices.

A key feature of the proposed system is its ability to automate complex tasks using natural language processing. To reduce the workload of instructors, the platform includes an AI-driven quiz generation feature that allows educators to upload PDF-based study materials. The system then analyzes the content using advanced language models, such as LLaMA, to extract key concepts and automatically generate multiple-choice questions. In addition, the platform enhances multimedia learning through automated video transcription powered by AI models like Groq Whisper. This functionality improves accessibility by enabling students to follow along with video lectures in text form and easily search for specific topics within the content.

Beyond automation, the system offers a structured and personalized learning experience through role-based dashboards. These dashboards provide real-time tracking of progress, assignments, and performance metrics. Learners benefit from an intuitive interface that displays completed modules and assessment results, while instructors have access to

centralized tools for course creation, content management, and automated grading. The integration of cloud storage services, such as Cloudinary, further supports efficient media handling and scalability.

2.3 FEASIBILITY STUDY

A feasibility study is conducted to evaluate the practicality and viability of the proposed AI-Powered E-Learning Platform from economic, operational, and technical perspectives. This analysis ensures that the project aligns with the objectives of EroSkillUp while remaining achievable within constraints such as time, cost, and available resources. The feasibility of the project is assessed across three key dimensions:

2.3.1 Economical Feasibility

The economical feasibility analysis focuses on comparing the financial investment required for development, deployment, and maintenance with the expected long-term benefits. The proposed system demonstrates strong economic viability due to the following factors:

- **Use of Open-Source Technologies:** The system is developed using the MERN stack (MongoDB, Express.js, React.js, Node.js), which is open-source and eliminates the need for costly software licensing.
- **Cost-Effective Infrastructure:** Cloud-based services such as MongoDB Atlas and Cloudinary are utilized for database and media management. These platforms provide scalable, pay-as-you-go pricing models along with free-tier options, reducing initial infrastructure costs.
- **Efficient AI Integration:** Instead of investing in expensive hardware such as GPU servers, the system integrates AI functionalities through external APIs (e.g., Groq SDK for LLaMA and Whisper models), significantly lowering computational expenses.
- **Improved Return on Investment (ROI):** Automation of tasks such as quiz generation and assignment evaluation reduces manual workload for instructors. This enables the institute to handle a larger number of students efficiently without increasing administrative costs.

2.3.2 Operational Feasibility

Operational feasibility evaluates how effectively the proposed system fits into the existing workflow and whether it addresses current challenges in the educational process. The system is highly feasible operationally due to the following reasons:

- **Automation of Educational Workflows:** Instructors can upload course materials such as PDFs, and the system automatically generates quizzes using AI, thereby reducing repetitive manual tasks and improving efficiency.
- **Enhanced Accessibility and Engagement:** Features such as automated video transcription support diverse learning needs, including accessibility for students with hearing impairments, and improve overall user engagement.
- **Centralized Monitoring and Analytics:** The platform provides comprehensive dashboards for administrators and instructors to track student progress, performance, and content engagement in real time, replacing fragmented monitoring systems.

2.3.3 Technical Feasibility

Technical feasibility assesses whether the required technologies, tools, and expertise are available to successfully develop and maintain the system. The proposed platform is technically feasible due to the following factors:

- **Reliable Technology Stack:** The MERN stack is a well-established and widely used development framework with strong community support, making it suitable for building scalable and reliable applications.
- **Efficient Backend Processing:** Node.js supports asynchronous and event-driven programming, enabling efficient handling of multiple concurrent operations such as API calls, media processing, and database interactions.
- **Scalable AI Integration:** Computationally intensive tasks such as natural language processing and audio transcription are handled through external AI services (e.g., Groq SDK), ensuring optimal system performance without overloading the server.

CHAPTER 3

SYSTEM SPECIFICATION

The system specification defines the technological requirements necessary for the development, testing, and deployment of the AI-Powered E-Learning Platform. The application is built using a modern web architecture and leverages cloud-based APIs, such as Groq, for handling computationally intensive AI operations. As a result, the system does not require high-end local hardware but instead depends on a reliable software environment and stable internet connectivity. The specifications are categorized into hardware and software requirements.

3.1 HARDWARE REQUIREMENTS

- **Processor**: Intel(R) Core(TM) i5 Processor / AMD Ryzen 5 (or equivalent)
- **Development**: Minimum 4 GB RAM, 10 GB available disk space (Local deployment)
- **Production**: 2 vCPU, 8 GB RAM (Cloud Hosting - Render/Vercel), highly scalable storage via Cloudinary for multimedia
- **Others**: High-speed Internet Connection (for API requests, cloud database access, and large multimedia uploads)

3.2 SOFTWARE REQUIREMENTS

- **Operating System**: Windows 10/11, macOS, or Linux (Ubuntu)
- **Core Platform**: MERN Stack (MongoDB, Express.js, React.js, Node.js)
- **Database**: MongoDB (Hosted via MongoDB Atlas cloud service)
- **Frontend UI/UX**: React.js, React Router v7, Framer Motion (Animations), Recharts (Data Visualization)
- **Cloud Services**: Cloudinary API (for scalable video, image, and PDF storage), Razorpay API (for payments)
- **AI Services**: Groq API SDK (Utilizing Whisper-large-v3 for video transcription)
- **Languages**: JavaScript (ES6+), JSX, HTML5, standard CSS3

CHAPTER 4

SOFTWARE DESCRIPTION

The software architecture of the AI-Powered E-Learning Platform is designed using a modern, multi-layered approach based on the MERN stack (MongoDB, Express.js, React.js, Node.js). This architecture is further enhanced through the integration of advanced artificial intelligence services and cloud-based APIs such as Groq (for Whisper and LLaMA models). The system follows a clear separation of concerns, dividing responsibilities among the presentation layer (frontend), application logic layer (backend), and data layer (database). This structured design improves scalability, maintainability, and security, while also ensuring high performance and responsiveness.

Additionally, the use of cloud services for AI processing and media storage significantly reduces the computational burden on local systems. This allows the platform to handle complex operations such as transcription and quiz generation efficiently without requiring specialized hardware.

4.1 FRONT END

The front end of the AI-Powered E-Learning Platform plays a crucial role in delivering an engaging, responsive, and user-centric experience. It acts as the primary interface through which all users—including learners, instructors, and administrators—interact with the system. The design philosophy of the frontend emphasizes usability, accessibility, responsiveness, and performance to ensure that users can seamlessly access educational content and system features across various devices.

The frontend is developed using React.js, a powerful and widely used JavaScript library for building dynamic user interfaces. React’s component-based architecture enables the application to be broken down into reusable, independent components such as navigation bars, dashboards, course cards, video players, and quiz modules. This modular approach not only improves code maintainability but also enhances development efficiency by allowing components to be reused across different parts of the application.

A key architectural feature of the frontend is its implementation as a Single Page Application (SPA). Unlike traditional multi-page applications, where each interaction requires a full page reload, the SPA dynamically updates only the necessary parts of the interface. This significantly reduces load times, minimizes server requests, and provides a smooth, uninterrupted user experience—especially important for e-learning platforms where users frequently switch between lessons, assessments, and dashboards. The frontend is also designed to be fully responsive, ensuring compatibility across desktops, tablets, and mobile devices. This responsiveness is achieved through modern CSS techniques and flexible layouts, allowing users to access the platform anytime and anywhere without compromising usability.

4.1.1 Features

The front-end architecture is enhanced with several advanced features and libraries that improve usability, performance, and visual appeal:

- **Efficient Rendering (Virtual DOM):** React updates only changed components, improving performance for dynamic features like progress tracking and AI transcripts.
- **Client-Side Routing (React Router):** Enables smooth navigation and restricts access based on user roles.
- **Data Visualization (Recharts):** Displays progress and performance data through clear and interactive charts.
- **Animations (Framer Motion):** Adds smooth transitions and effects for a better user experience.
- **Real-Time Notifications (React Hot Toast):** Provides instant, non-intrusive feedback for user actions.

4.2 BACK END

The backend of the AI-Powered E-Learning Platform functions as the central processing unit, managing application logic, handling client requests, processing data, and coordinating communication across system components.

Built using Node.js, it leverages an event-driven, non-blocking I/O model that enables efficient handling of multiple concurrent user requestsan essential feature for an e-learning system with simultaneous activities such as content access, assignment submissions, and AI interactions.

The platform also utilizes Express.js, a lightweight web framework that simplifies the development of RESTful APIs through structured routing, middleware support, and efficient request handling. This allows the system to manage key functionalities like authentication, course management, file handling, and AI operations in a scalable and organized manner. Additionally, the backend integrates with external AI services, such as Groq, to perform tasks like quiz generation and video transcription, reducing server load while delivering advanced intelligent features.

The backend is also responsible for file management, cloud integration, and system security. It processes user-uploaded files using middleware and stores them in cloud services like Cloudinary, ensuring efficient handling of large multimedia content such as videos and documents. In terms of security, the system implements robust mechanisms including password hashing with bcryptjs and session management using JSON Web Tokens (JWT). These measures support secure authentication and enable role-based access control, ensuring that only authorized users—whether administrators, instructors, or students—can access specific features of the platform.

4.2.1 Features

The backend system incorporates several advanced capabilities to ensure performance, scalability, and security:

- **Asynchronous and Concurrent Processing:** Node.js allows the system to handle multiple requests simultaneously without blocking execution. This is particularly useful for operations such as uploading large files, processing AI requests, and handling multiple user interactions at the same time.

- **AI Integration and Automation:** The backend integrates with AI services through APIs to perform tasks such as generating quizzes from PDF documents and transcribing video content. This reduces manual effort and enhances the overall functionality of the platform.

- **Cloud-Based Media Management:** File uploads are handled using middleware and directly stored in cloud platforms like Cloudinary. This ensures efficient storage, fast retrieval, and scalability for handling large volumes of multimedia content.

- **Secure Authentication and Authorization:** The system uses bcryptjs for encrypting user passwords and JSON Web Tokens (JWT) for managing secure user sessions. Role-based access control ensures that users can only access features relevant to their roles.

CHAPTER 5

PROJECT DESCRIPTION

5.1 PROBLEM DEFINITION

Over the past decade, digital education has grown rapidly, yet many existing Learning Management Systems (LMS) still rely on outdated approaches. These platforms primarily act as static content repositories with limited interactivity and adaptability, making them insufficient for modern educational needs. A major challenge is the inefficiency of manual curriculum management, where instructors must create quizzes, design assessments, and evaluate submissions manually. This process is time-consuming, often inconsistent, and reduces the time available for meaningful teaching and student engagement. Additionally, delayed feedback in such systems prevents learners from identifying and correcting mistakes in real time, disrupting the continuous learning process and slowing overall progress.

Another significant limitation is the lack of personalized learning experiences. Traditional LMS platforms adopt a one-size-fits-all approach, delivering the same content at the same pace to all learners, without considering individual differences in learning ability, prior knowledge, or comprehension levels. This results in disengagement among advanced learners and difficulties for those who need additional support.

Furthermore, the limited accessibility and searchability of multimedia content, especially video-based materials, create additional barriers. The absence of automated transcription and indexing makes it difficult for students to locate specific information and poses challenges for hearing-impaired learners. Overall, the core issue lies in the lack of intelligent automation and adaptability, as current systems fail to leverage artificial intelligence to provide personalized, efficient, and responsive learning experiences.

5.2 OVERVIEW OF THE PROJECT

The AI-Powered E-Learning Platform is developed to address the shortcomings of traditional digital learning systems by embedding advanced artificial intelligence directly into the educational process. It transforms conventional LMS platforms into dynamic, interactive, and personalized environments that meet the needs of learners, instructors, and administrators. By leveraging Natural Language Processing (NLP) and machine learning techniques through the Groq SDK, the platform automates complex academic tasks, significantly reducing manual effort and improving overall efficiency. Key features include intelligent quiz generation, where instructors can upload study materials such as PDFs and the system automatically extracts important concepts to generate relevant multiple-choice questions, ensuring alignment with course content while saving valuable time.

Another essential capability is automated video transcription, which enhances accessibility and usability in multimedia learning. The system converts uploaded video lectures into accurate, timestamped transcripts, allowing students to follow along, search for specific topics, and benefit from improved accessibility, especially for differently-abled learners. Additionally, the platform offers role-based dashboards that create a structured and customized user experience: learners can access content, track progress, and receive instant feedback; instructors can manage courses and monitor performance; and administrators can oversee system operations, analytics, and financial processes. Built using the MERN stack, the platform ensures scalability and performance, with a React-based frontend for a seamless interface, a Node.js and Express.js backend for efficient processing, and MongoDB for flexible data management. Integration with services like Cloudinary for media storage and Razorpay for secure payments further strengthens the platform’s functionality and reliability.

5.3 MODULE DESCRIPTION

The AI-Powered E-Learning Platform is organized into five major functional modules. Each module is responsible for a specific area of the application's functionality, and together they form a complete, intelligent online learning ecosystem. The following sections provide a detailed description of each module along with its key features.

5.3.1 User & Admin Management

The User and Admin Management module serves as the foundation of the platform by managing user identities and controlling access based on roles. It supports three types of users Students, Instructors, and Administrators each with specific permissions and dedicated interfaces tailored to their needs. New users can register by providing basic details such as name, email, and password, which is securely encrypted before being stored in the database to ensure data protection.

Upon login, the system verifies credentials, identifies the user's role, and redirects them to the appropriate dashboard. A key feature of this module is Role-Based Access Control, which restricts users from accessing functionalities outside their permissions, ensuring security and system integrity through both interface-level restrictions and server-side validation.

Key Features:

- **Secure Account Registration and Login:** Users can create accounts with encrypted password storage. The login process seamlessly identifies the user's role and directs them to the appropriate portal—whether it is the Learner Dashboard, the Instructor Dashboard, or the Admin Control Panel.

- **Comprehensive Profile Management:** Every user can personalize their profile by updating their name, phone number, location, biography, and links to their social media profiles such as LinkedIn, Twitter, and YouTube. Instructors can additionally update their professional title and area of expertise.

- **Resume Upload:** Students and professionals can upload their resumes in PDF format. The file is securely stored in the cloud and linked to their profile, making it accessible for career development purposes.

- **Administrative User Oversight:** The Administrator has a dedicated panel that provides a bird's-eye view of all registered users on the platform. This includes the ability to view user details, monitor registration activity, and manage instructor accounts.

5.3.2 Course & Content Management

The Course and Content Management module acts as the operational core of the platform, enabling the creation, organization, delivery, and management of educational content while offering students a smooth browsing and enrollment experience. Instructors and administrators can create courses by providing details such as title, description, category, price, and a thumbnail image, and then enrich them with multimedia content including video lectures, PDFs, and images, all securely stored in the cloud for reliable access.

From the learner's perspective, the platform functions as a digital marketplace where students can explore courses, view details like instructor information, ratings, and enrollment count, and purchase courses through a streamlined cart and checkout process. Once enrolled, students gain instant access to course materials, while an integrated notification system keeps them informed about new courses and content updates, ensuring continuous engagement.

Key Features:

- **Complete Course Lifecycle Management:** Instructors and administrators can create new courses, update existing course details and thumbnails, and delete courses that are no longer relevant. Deleting a course also cleans up all associated media files from the cloud storage.

- **Diverse Multimedia Content Support:** Each course can contain an unlimited mix of video lectures, PDF study materials, and image resources. Content can be added, edited, or removed at any time, and instructors can provide titles and descriptions for each item.

- **Student Shopping Cart & Checkout:** A fully functional shopping cart allows students to add multiple courses, review a detailed order summary including subtotals and pricing, apply coupon codes, and complete their purchase in a streamlined checkout flow.

- **Real-Time Student Enrollment Tracking:** Each course dynamically displays the number of currently enrolled students, providing social proof and helping new learners gauge the popularity and relevance of a course.

5.3.3 AI-Powered Personalization

The AI-Powered Personalization module is the key innovation of the platform, transforming it from a traditional content delivery system into an intelligent and adaptive learning environment by integrating advanced AI capabilities. It automates complex tasks such as video transcription, where uploaded lectures are converted into accurate, timestamped text displayed alongside videos with synchronized highlighting, enhancing accessibility for hearing-impaired learners and those who prefer reading. Another major feature is intelligent quiz generation, which allows instructors to upload study materials like PDFs, enabling the system to analyze content, extract key concepts, and automatically generate structured multiple-choice questions with correct answers, saving time and ensuring relevance. Additionally, the module includes a learning path personalization feature, where students provide their interests, career goals, and desired skills through a guided setup, allowing the platform to recommend tailored courses and resources, ultimately delivering a customized, engaging, and efficient learning experience.

Key Features:

- **Real-Time Transcript Synchronization:** As a video plays, the corresponding text is highlighted in real-time and the view automatically scrolls to keep pace, creating an interactive read-along experience that enhances comprehension and retention.

- **AI-Powered Smart Search in Videos:** Students can type any keyword into a search bar, and the system instantly finds all moments in the video where that topic is discussed. Clicking on a search result jumps the video directly to the relevant timestamp, saving students significant time when revising specific topics.

- **Personalized Learning Path Wizard:** A guided, three-step personalization questionnaire captures each student's professional field, target occupation, and skill interests. This information is stored and used to surface the most relevant courses and learning recommendations for each individual.

- **Automatic Quiz Generation from PDFs:** Instructors upload a PDF study guide, and the AI intelligently reads the content and generates a ready-to-use quiz with ten multiple-choice questions. This drastically reduces the preparation time required for assessments.

5.3.4 Assessment & Analytics

The Assessment and Analytics module provides a complete system for evaluating student performance and delivering meaningful insights to both learners and instructors by managing the entire assessment process, including assignment and quiz creation, submissions, grading, and result visualization through user-friendly dashboards. Instructors can create assignments with instructions and deadlines, where students submit files securely, as well as interactive quizzes that offer instant feedback with scores and correct answers. The module also features automated progress tracking, continuously monitoring completed content, assignment submissions, and quiz performance (with an 80% passing benchmark) to calculate real-time course progress displayed via progress bars. Additionally, a centralized gradebook allows instructors to view all student data in one place, manually grade submissions when required, and identify learners who need additional support.

Key Features:

- **Interactive Quiz Taking and Instant Scoring:** Students take quizzes in an engaging, interactive format within the course viewer. Upon submission, they receive an immediate score along with a detailed review showing correct answers versus their selections.

- **Assignment Submission via File Upload:** Students can submit assignment work by uploading files directly through the platform. The system records the submission timestamp, filename, and cloud storage link for the instructor to review.

- **Real-Time Automated Progress Tracking:** The platform dynamically calculates each student's completion percentage based on their completed content, submitted assignments, and passed quizzes, presenting this as a visual progress bar that updates in real-time.

- **Instructor Gradebook with Manual Grading:** Instructors have access to a centralized gradebook that lists all enrolled students and their performance across every assessment. Manual grading is supported for assignments that require subjective evaluation.

5.3.5 Interactive Learning & NLP Chatbot

The Interactive Learning and NLP Chatbot module enhances the learning experience by making it more engaging, accessible, and interactive through the integration of intelligent features and communication tools. At its core is an advanced content viewer that organizes course materials into categories such as video lessons, study materials, assignments, and quizzes, with clear indicators for completion and easy navigation. It also includes an NLP-powered transcript and search system that converts video lectures into searchable content, allowing students to find specific topics instantly and jump to relevant timestamps. Additionally, the module features a robust notification and communication system that keeps users informed about course updates, new materials, and deadlines, along with a Contact and Support feature for submitting queries or feedback. Together, these elements create a dynamic, user-friendly environment that keeps students actively engaged and well-informed throughout their learning journey.

Key Features:

1. **Intelligent NLP Chatbot Assistant:** A 24/7 virtual teaching assistant that interprets natural language queries, allowing students to ask conceptual questions and receive instant, context-aware answers regarding course materials.

2. **Custom Interactive Video Player:** A highly responsive video player featuring advanced learning controls, including precision 10-second skip forward/backward functions via on-screen buttons or keyboard shortcuts.

3. **Real-Time Contextual Notifications:** An automated alert system that pushes vital updates directly to the student's dashboard, ensuring they never miss upcoming assignment deadlines or newly published course materials.

4. **Contact & Support Portal:** A dedicated support page allows students to submit structured queries with their name, email, subject, and message. All submissions are stored and made available to administrators for timely review and resolution.

5.4 PROCESS FLOW DIAGRAM

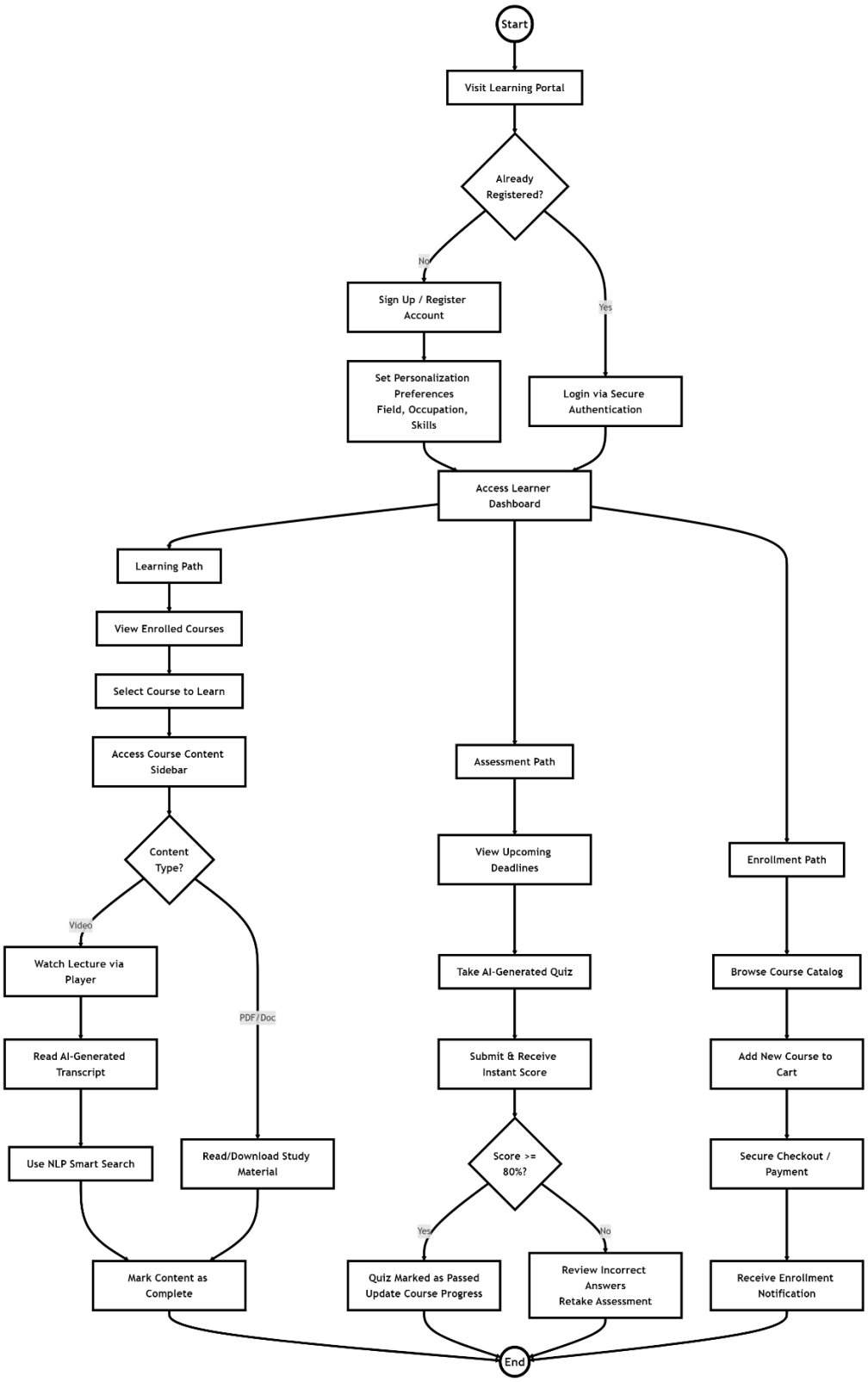

Figure 5.4.1 Learner Process Flow

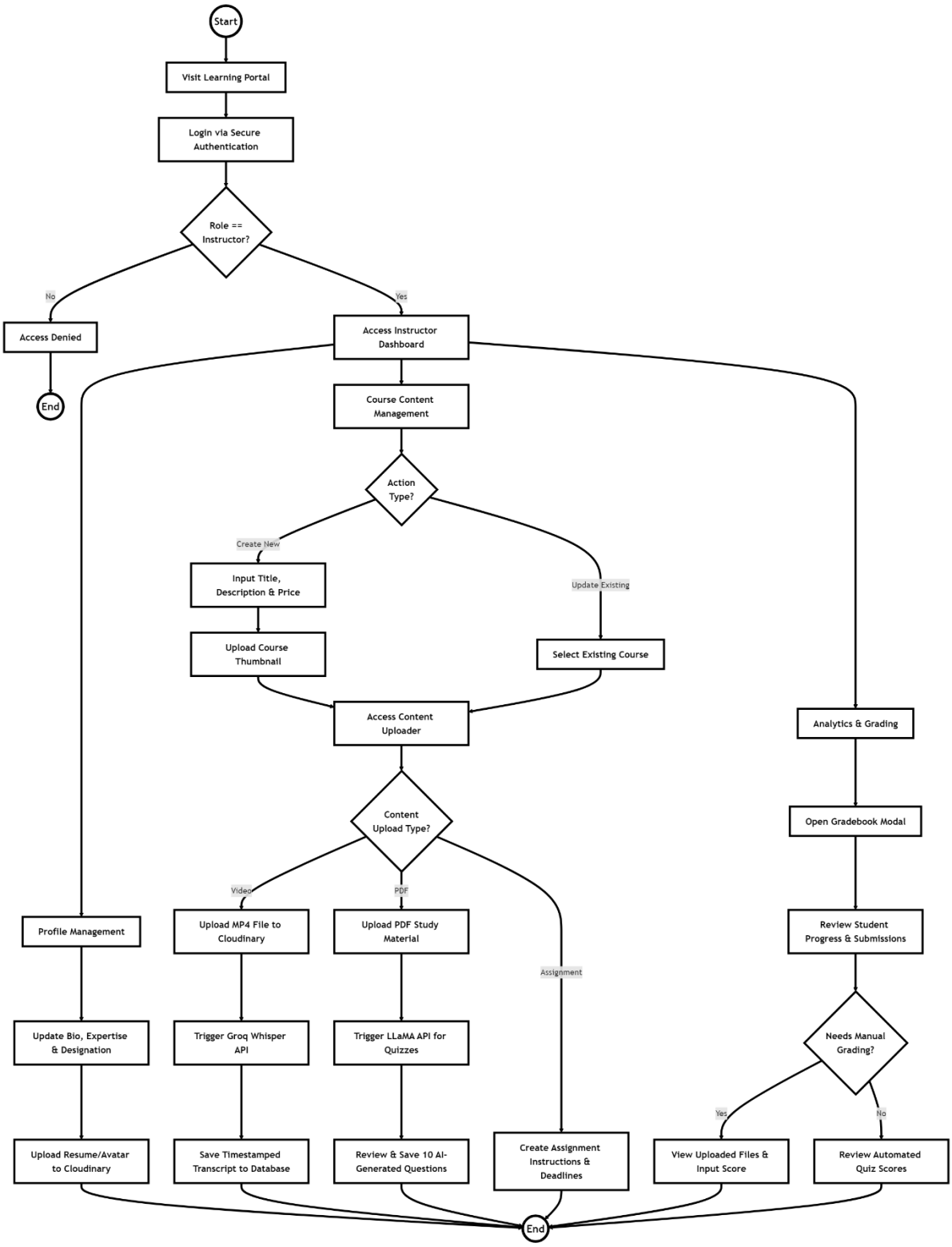

Figure 5.4.2 Instructor Flow Diagram

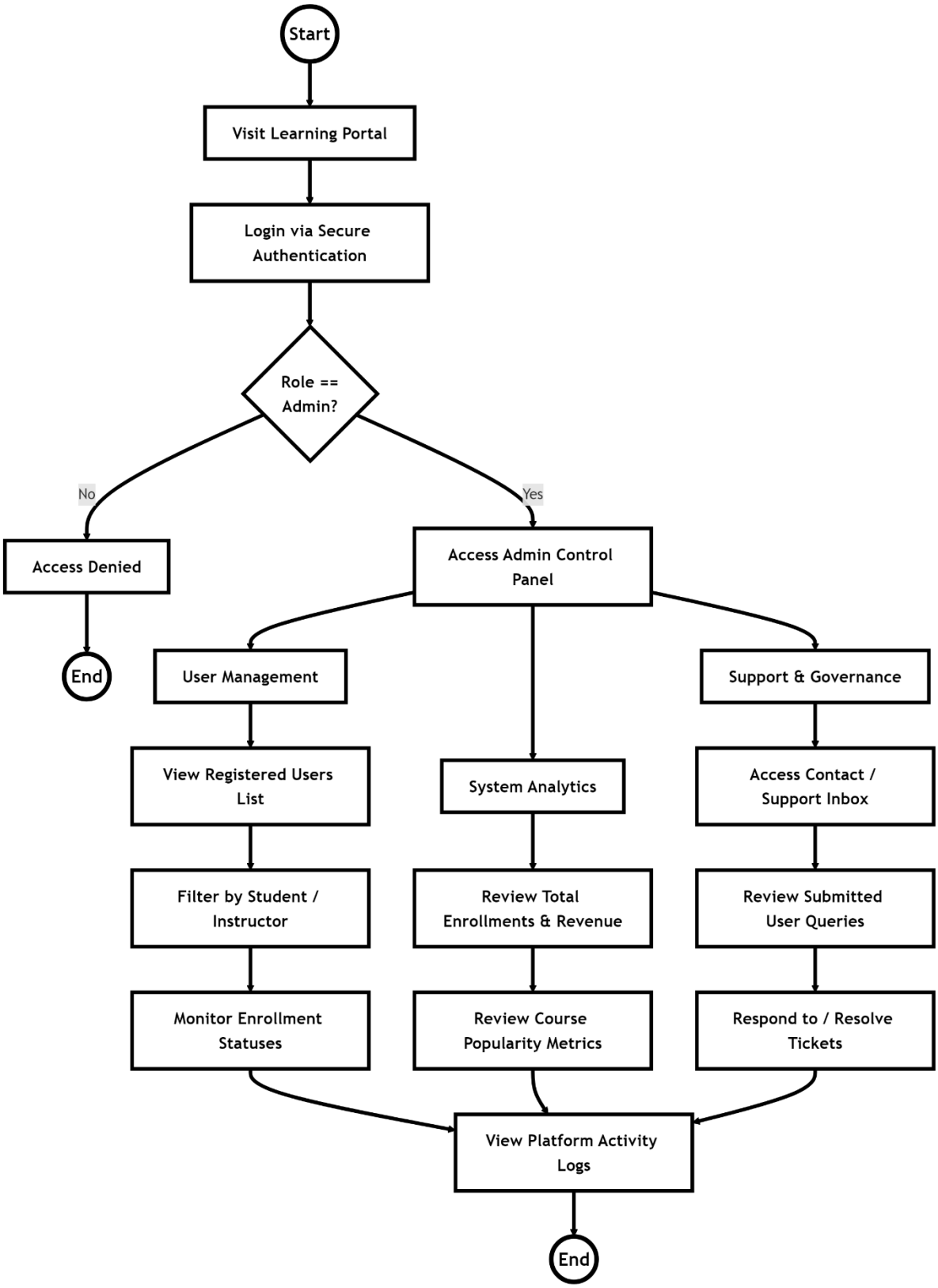

Figure 5.4.3 Admin Flow Diagram

5.5 DATA FLOW DIAGRAM

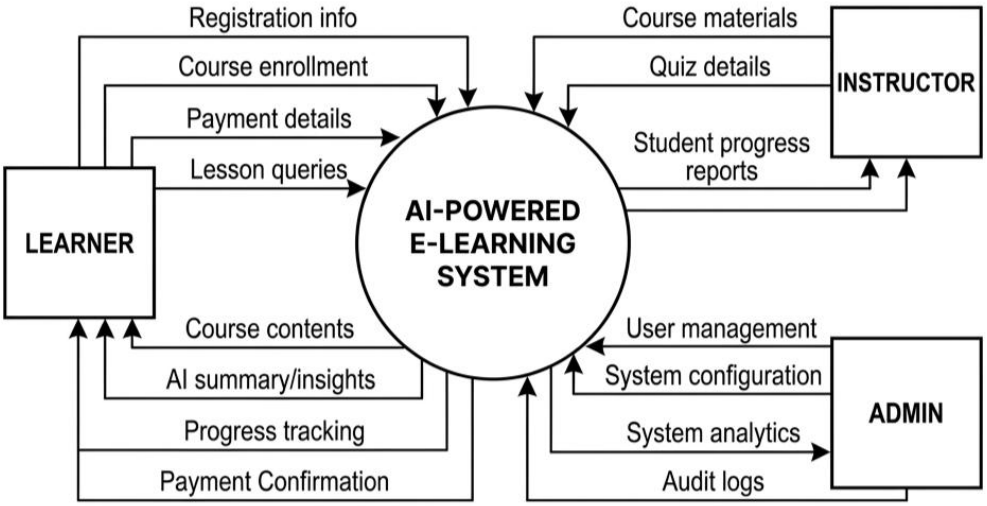

Figure 5.5.1 Data Flow Diagram Level 0

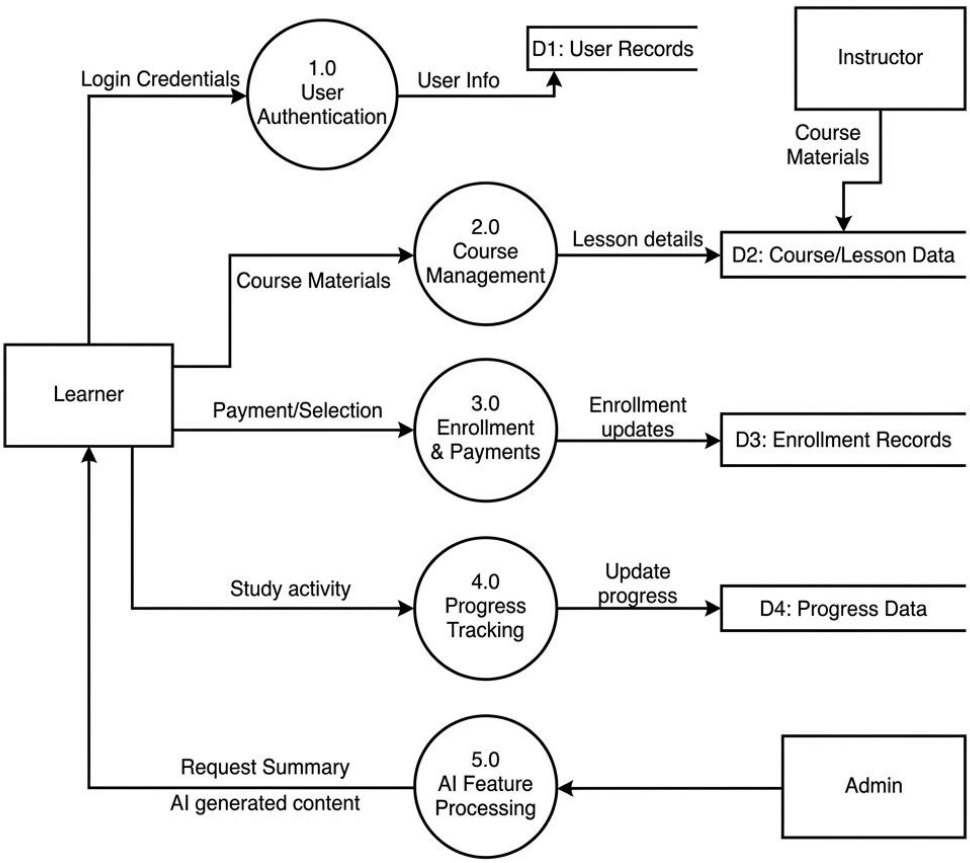

Figure 5.5.2 Data Flow Diagram Level 1

5.6 DATABASE DESIGN

The database forms the backbone of the AI-Powered E-Learning Platform, enabling the persistent storage of user data, course content, assessments, progress tracking, and system logs. The platform utilizes MongoDB, a NoSQL database that stores data in a flexible, document-oriented format similar to JSON, allowing it to efficiently handle varying course structures such as video-heavy or document-based content. To maintain consistency within this flexible system, Mongoose, an Object Data Modeling (ODM) library, is used to define schemas that act as blueprints for each document, specifying required fields, data types, and validation rules. This ensures that the data remains structured, reliable, and predictable. Overall, the database is organized into seven distinct collections, each designed to manage specific aspects of the platform's functionality.

Table 5.6.1 Admin Table

Field Name	Description	JSON
id	Unique identifier for the admin account	{ "_id": "64f6c3d4e5f6g7h8i9j0k1l2", "name": "Super Admin", "email": "admin@learningapp.com", "role": "admin", "location": "Global Operations Center", "socialLinks": { "linkedin": "https://linkedin.com/in/admin-root", "github": "https://github.com/admin-root" }, "createdAt": "2024-01-01T09:00:00Z" }
name	Full name of the administrator (required)	
email	Email address, must be unique (required)	
password	Account password (required)	
role	Always set to 'admin'	
phone	Contact phone number	
location	City or region	
bio	Short biography	
resume	Cloud storage URL of the admin's resume	
socialLinks	Links to social media profiles	
createdAt	Account creation date	

Table 5.6.2 Instructor Table

Field Name	Description	JSON
id	Unique identifier for the instructor profile	{ "_id": "64f3c2a1b0c1234567890abc", "name": "Dr. Harrison Smith", "email": "h.smith@learningapp.com", "designation": "Head of Data Science", "bio": "Experienced researcher with 10+ years in distributed systems.", "expertise": ["MongoDB", "Big Data", "System Design"], "image": "https://cloudinary.com/profiles/hsmith.jpg" }
name	Full name of the instructor (required)	
email	Email address, must be unique (required)	
designation	Professional title (required)	
bio	A detailed biography of the instructor (required)	
expertise	A list of the instructor's skills (required)	
image	Cloud storage URL of the instructor's profile photograph	
createdAt	The date the instructor profile was created	

Table 5.6.3 Contact Table

Field Name	Description	JSON
id	Unique identifier for the contact message	{ "_id": "64f5b2...", "name": "Alex", "email": "a@mail.com", "subject": "Hi", "message": "Hello", "createdAt": "2024-04-18"}
name	The name of the person submitting the message	
email	The email address of the sender	
subject	The subject or topic of the inquiry	
message	The detailed message body	
createdAt	The date and time when the message was submitted	

Table 5.6.4 Notification Table

Field Name	Description	JSON
id	Unique identifier for the notification	{
userId	A reference to the user who should receive this notification	"_id": "64f4a1...",
title	A brief heading for the notification (required)	"userId": "64f3a8...",
message	The detailed notification message (required)	"title": "Alert",
type	The category of notification	"message": "Msg",
unread	Indicates whether the user has read the notification (defaults to true/unread)	"type": "system",
createdAt	The date and time when the notification was generated	"unread": true,
		"createdAt": "2024-04-18"
		}

Table 5.6.5 Activity Log Table

Field Name	Description	JSON
id	Unique identifier for the log entry	{
action	The type of action performe	"_id": "64f3f9...",
description	A human-readable description of what happened (required)	"action": "Login",
entityId	The identifier of the entity that was affected	"description": "User login",
entityName	The name of the affected entity for easy identification	"entityId": "U1",
createdAt	The exact date and time when the action occurred	"entityName": "User",
		"createdAt": "2024-04-18"
		}

Table 5.6.6 User Table

Field Name	Description	JSON
id	Unique identifier automatically generated by MongoDB	{ "_id": "64f3a8b0e4b0c12345678901", "name": "Jane Student", "email": "jane.student@example.com", "password": "$2b$10$hashed_password_string", "role": "student", "enrolledCourses": [{ "courseId": "64f3d1b25f1b2c4d5e6f7g8h", "progress": 75, "completedContent": ["vid_001", "pdf_002"] }], "preferences": { "field": "Web Development", "skills": ["JavaScript", "HTML"] }, "createdAt": "2024-04-18T10:00:00Z" }
name	Full name of the user (required)	
email	Email address, must be unique across all users (required)	
phone	Contact phone number	
location	City or region of the user	
password	Encrypted password stored in hashed format (required)	
role	Defines the user's role — can be 'student', 'instructor', or 'admin'	
enrolledCourses	A list of all courses the user is enrolled in (detailed below)	
cart	A list of course IDs the user has added to their shopping cart	
preferences	Stores the user's personalization settings — field, occupation, and skills	
bio	A short biography written by the user	
resume	Cloud storage URL of the user's uploaded resume	
socialLinks	Links to the user's social profiles — website, LinkedIn, Twitter, etc.	
createdAt	The date and time when the account was created	

Table 5.6.7 Course Table

Field Name	Description	JSON
id	Unique identifier for the course	{
title	The name of the course (required)	"_id": "64f3d1...",
description	A detailed description of what the course covers (required)	"title": "MongoDB",
instructor	The name of the instructor who teaches the course (required)	"description": "Learn NoSQL",
instructorId	A reference linking to the instructor's account in the User collection	"instructor": "Dr. Smith",
Sprice	The cost of the course in the platform's currency (required)	"instructorId": "64f3c2...",
rating	The average rating given by students (defaults to 0)	"price": 49,
category	The subject category the course belongs to (required)	"rating": 4.5,
thumbnail	Cloud storage URL of the course's cover image	"numReviews": 10,
lessons	A list of lesson objects, each containing a title, content description, and duration	"reviews": [{ "user": "ID", "name": "Jane", "rating": 5, "comment": "Good" }],
content	A list of all uploaded multimedia content items (detailed below)	"category": "DB",
assignments	A list of all assignments created for this course	"thumbnail": "img.jpg",
quizzes	A list of all quizzes created for this course	"lessons": [{ "title": "Intro", "duration": 10 }],
createdAt	The date the course was created	"content": [{ "title": "Vid1", "type": "video", "url": "url" }],
		"assignments": [{ "title": "A1", "description": "Task" }],
		"quizzes": [{ "title": "Q1", "questions": [{"question": "Q?", "options": ["A"], "correctAnswer": 0}] }],
		"createdAt": "2024-04-18"
		}

CHAPTER 6

SYSTEM TESTING

System testing is a critical phase in the software development lifecycle that ensures the developed application functions correctly, reliably, and as intended before it is deployed for real-world use. As part of the system testing process for the AI-Powered E-Learning Platform, a systematic approach was followed to evaluate the accuracy, performance, and robustness of all features across the five major modules. The testing process was carried out at two levels—Unit Testing and Acceptance Testing—followed by the documentation of detailed test cases and their results.

6.1 UNIT TESTING

Unit testing is an essential aspect of the development process for the AI-Powered E-Learning Platform. It involves testing individual components or "units" of code in isolation to ensure they function correctly and reliably.

Thorough unit testing was conducted for all major modules: User Authentication, Course Management, AI-Powered Personalization, Assessment & Analytics, and the Learner Dashboard. Test cases encompass various scenarios such as successful user registration, encryption of passwords, error handling for missing mandatory fields, and validation of course upload formats. Special emphasis was placed on testing the AI service endpoints independently to ensure the Groq Whisper API correctly returns timestamped text for audio inputs, and the LLaMA model accurately extracts multiple-choice questions from provided text. Through comprehensive unit testing, the platform ensures that individual functions—like cart total calculation, progress bar rendering, or secure token generation—operate flawlessly before they are combined into larger system flows.

6.2 ACCEPTANCE TESTING

Acceptance testing (also known as User Acceptance Testing or UAT) evaluates the system's compliance with the business requirements and assesses whether it is acceptable for delivery to the end users. This testing checks the complete, integrated system from the perspective of the actual users: Students, Instructors, and Administrators.

In this phase, the application was tested against real-world scenarios to ensure smooth user journeys. For students, the testing verified that they could successfully browse the course catalog, add courses to their cart, complete the secure checkout process, navigate the video player, interact with the AI-generated transcripts, and submit quizzes. For instructors, the acceptance testing validated the flow of creating a new course, uploading massive video files and PDFs to Cloudinary, triggering the AI models for automated quiz and transcript generation, and reviewing the gradebook. For administrators, testing verified the ability to monitor platform-wide user analytics and handle contact support messages. Successful acceptance testing guaranteed that the AI-Powered E-Learning Platform meets the diverse and practical needs of all its target audiences.

Furthermore, Acceptance Testing served as the ultimate validation checkpoint for the platform’s core artificial intelligence features and its overall responsiveness. Test scenarios were executed across multiple devices—including desktop computers, tablets, and mobile phones—to guarantee that the fluid, single-page application experience and complex visual elements, like the custom video player, remained fully accessible and responsive under varying network conditions. Critically, stakeholders evaluated the subjective quality of the AI integrations. Similarly, the Groq Whisper transcriptions were scrutinized to ensure their accuracy and timestamp synchronization were of high enough quality to genuinely aid a student's learning process. By passing these rigorous real-world evaluations, the platform demonstrated it was robust, user-friendly, and fully prepared for deployment.

6.3 TEST CASES

The following table documents the critical test cases executed to ensure the platform's stability, security, and functionality.

TESTCASE ID	TESTCASE DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	RESULT
T1	If a user tries to register using an email address that already exists in the database.	Display error message "User already exists. Please login."	Display error message "User already exists. Please login."	Success
T2	If a registered Student attempts to access the Instructor or Admin Dashboard URL directly.	Access is denied, and the user is redirected back to the secure Learner Dashboard.	Access is denied, and the user is redirected back to the secure Learner Dashboard.	Success
T3	If an instructor uploads a valid PDF document and triggers the "Generate AI Quiz" function.	The AI model processes the text and successfully returns 10 multiple choice questions with correct answers.	The AI model processes the text and successfully returns 10 multiple choice questions with correct answers.	Success
T4	If an instructor uploads a video lecture and requests "AI Transcription".	The Whisper AI model generates and displays an accurate, timestamped transcript synced with the video.	The Whisper AI model generates and displays an accurate, timestamped transcript synced with the video.	Success
T5	If a student submits a course quiz and scores greater than or equal to 80%.	The quiz is marked as "Passed" and the student's overall course progress percentage increases.	The quiz is marked as "Passed" and the student's overall course progress percentage increases.	Success

T6	If a user attempts to submit the login form with empty email or password fields.	Display error message "Data cannot be empty" or "Please fill in all fields".	Display error message "Data cannot be empty" or "Please fill in all fields".	Success
T7	If a student clicks the "Checkout" button while their shopping cart is completely empty.	Display a warning message "Your cart is empty" and disable checkout progression.	Display a warning message "Your cart is empty" and disable checkout progression.	Success
T8	If a student types a specific topic (e.g., "React Hooks") into the Smart Video Transcript Search bar.	The system highlights and displays all transcript segments where the keyword was spoken, with clickable timestamps.	The system highlights and displays all transcript segments where the keyword was spoken, with clickable timestamps.	Success
T9	If a student clicks the "Mark as Complete" toggle on a specific PDF study material.	The material is instantly visually marked complete, and the progress bar updates successfully.	The material is instantly visually marked complete, and the progress bar updates successfully.	Success
T10	If a user updates their Personalization preferences (Field, Occupation, Skills) and saves.	Display success toast "Profile updated" and immediately reflect the new skills on their dashboard.	Display success toast "Profile updated" and immediately reflect the new skills on their dashboard.	Success

Table 6.3.1 Test Cases for AI-Powered E-Learning Platfor

CHAPTER 7

SYSTEM IMPLEMENTATION

System implementation is the phase in which the designed system architecture is converted into a fully functional and deployable application that can be accessed by users over the internet. This stage involves setting up the development environment, configuring cloud services, integrating third-party APIs, and deploying the application using cloud platforms such as Render and Vercel. The following steps outline the implementation process of the AI-Powered E-Learning Platform.

Cloud Database Setup (MongoDB Atlas)

- **Cluster Creation:** A cloud-based NoSQL database cluster is created using MongoDB Atlas to store application data securely and efficiently.
- **Database Connection:** The MongoDB connection string (URI) is obtained from the Atlas dashboard and used by the backend server to establish a secure connection with the database.

Third-Party API Configuration

- **AI Integration (Groq SDK):** API keys are generated from the Groq platform to enable access to AI models such as Whisper for video transcription and LLaMA for quiz generation.
- **Media Storage (Cloudinary):** A Cloudinary account is configured to manage image and video uploads. API credentials are integrated into the application to ensure efficient storage and delivery of media content through CDN services.

CHAPTER 8

CONCLUSION & FUTURE ENHANCEMENTS

8.1 CONCLUSION

The development and implementation of the AI-Powered E-Learning Platform represents a significant leap forward in modernizing digital education by seamlessly integrating advanced artificial intelligence into a robust, full-stack web architecture. By transitioning away from rigid, traditional Learning Management Systems, this project successfully resolved several systemic inefficiencies—specifically the heavy administrative burden on instructors and the lack of accessible, personalized learning tools for students. Utilizing the MERN stack alongside powerful integrations like the Groq Whisper model for automated, timestamped video transcriptions and the LLaMA versatile model for instant quiz generation, the platform proved that complex educational tasks can be highly automated without sacrificing quality. This intelligent automation empowers educators to shift their invaluable time away from manual data entry and toward active student mentoring and advanced curriculum design.

Furthermore, by transforming raw video audio into fully searchable, NLP-driven text segments, the platform drastically improved accessibility for hearing-impaired learners and revolutionized how all students search for and review specific lecture topics. The inclusion of customized role-based dashboards, secure financial integrations via Razorpay, resilient Cloudinary multimedia storage, and dynamic progress analytics guarantees a highly responsive, user-centric environment that scales efficiently. Ultimately, the successful deployment of this cloud-native platform validates the immense potential of AI to transform online learning from a passive, one-size-fits-all experience into an adaptive, inclusive, and highly engaging interactive educational ecosystem built for the modern era.

8.2 FUTURE ENHANCEMENTS

- **AI-Powered Adaptive Learning Engine:** Implement a deep learning system that analyzes each student's performance patterns and dynamically generates a fully personalized learning path—automatically reordering modules and adjusting difficulty levels in real-time.

- **Real-Time Collaborative Virtual Classroom:** Integrate live video conferencing, shared whiteboards, and synchronized code editors directly within the platform, enabling instructors to conduct live lectures and peer-to-peer collaboration sessions.

- **AI-Based Plagiarism Detection & Essay Grading:** Develop an NLP-powered engine that detects plagiarism across student submissions and automatically evaluates free-text answers based on relevance and coherence, reducing manual grading workload.

- **Multi-Language Support with AI Translation:** Implement real-time AI-powered translation of all course content—video transcripts, quizzes, and UI elements—into the student's preferred language, making the platform globally accessible.

- **AI-Powered Visual Content Generation:** Implement an AI system that automatically generates visual aids—such as diagrams, mind maps, and infographics—from PDF study materials and video transcripts, helping visual learners grasp complex concepts more effsectively.

CHAPTER 9

APPENDIX

9.1 SOURCE CODE

App.jsx

```
import { BrowserRouter as Router, Routes, Route, useLocation } from 'react-router-dom';

import { Toaster } from 'react-hot-toast';

import ScrollToTop from './components/Common/ScrollToTop';

import Navbar from './components/Common/Navbar';

import Footer from './components/Common/Footer'; // Import Footer

import Home from './pages/Learner/Home';

import Login from './pages/Auth/Login';

import Signup from './pages/Auth/Signup';

import Dashboard from './pages/Learner/Dashboard';

import UserProfile from './pages/Learner/UserProfile';

import Courses from './pages/Learner/Courses';

import CourseOverview from './pages/Learner/CourseOverview';

import CourseContent from './pages/Learner/CourseContent';

import About from './pages/Learner/About';

import Contact from './pages/Learner/Contact';

import Cart from './pages/Learner/Cart';

import AdminDashboard from './pages/Admin/AdminDashboard';

import AdminUsers from './pages/Admin/AdminUsers';

import AdminSettings from './pages/Admin/AdminSettings';
```

```
import AdminInstructors from './pages/Admin/AdminInstructors';

import AdminProfile from './pages/Admin/AdminProfile';

import InstructorDashboard from './pages/Instructor/InstructorDashboard';

import InstructorCourses from './pages/Instructor/InstructorCourses';

import InstructorStudents from './pages/Instructor/InstructorStudents';

import AdminEarnings from './pages/Admin/AdminEarnings';

import InstructorProfile from './pages/Instructor/InstructorProfile';

import InstructorSettings from './pages/Instructor/InstructorSettings';

import InstructorCourseDetails from './pages/Instructor/InstructorCourseDetails';

import Chatbot from './components/Chatbot/Chatbot';

const NavbarWrapper = () => {

  const location = useLocation();

  const isAdminRoute = location.pathname.startsWith('/admin') ||
location.pathname.startsWith('/instructor');

  if (isAdminRoute) return null;

  return <Navbar />;};

const FooterWrapper = () => { // Create Footer Wrapper

  const location = useLocation();

  const isAdminRoute = location.pathname.startsWith('/admin') ||
location.pathname.startsWith('/instructor');

  const isCourseContent = location.pathname.startsWith('/course-content');

  if (isAdminRoute || isCourseContent) return null;

  return <Footer />;};

const ChatbotWrapper = () => {

  const location = useLocation();

  const isAdminRoute = location.pathname.startsWith('/admin') ||
location.pathname.startsWith('/instructor');

  if (isAdminRoute) return null;
```

```
  return <Chatbot />;};

function App() {

  return (<Router>

      <ScrollToTop />

      <Toaster position="top-center" toastOptions={{

        style: {

          background: '#333',

          color: '#fff', },

        success: {

          style: {

            background: '#10b981', },},

        error: {

          style: {

            background: '#ef4444', },},}}/>

      <div className="App" style={{ minHeight: '100vh', display: 'flex', flexDirection: 'column' }}>

        <NavbarWrapper />

        <main style={{ flex: 1, paddingTop: '0' }}> {/* Removed padding for admin overlap control in
their layout */}

          <Routes>

            <Route path="/" element={<div style={{paddingTop: '80px'}}><Home /></div>} />

            <Route path="/login" element={<div style={{paddingTop: '80px'}}><Login /></div>} />

            <Route path="/signup" element={<div style={{paddingTop: '80px'}}><Signup /></div>} />

            <Route path="/dashboard" element={<div style={{paddingTop: '80px'}}><Dashboard
/></div>} />

          <Route path="/profile" element={<div style={{paddingTop: '0px'}}><UserProfile /></div>} />

            <Route path="/courses" element={<div style={{paddingTop: '80px'}}><Courses /></div>} />

            <Route path="/course/:id" element={<div style={{paddingTop: '80px'}}><CourseOverview
/></div>} />
```

```
          <Route path="/course-content/:id" element={<div style={{paddingTop:
'0px'}}><CourseContent /></div>} />

          <Route path="/about" element={<div style={{paddingTop: '80px'}}><About /></div>} />

          <Route path="/contact" element={<div style={{paddingTop: '80px'}}><Contact /></div>} />

          <Route path="/cart" element={<div style={{paddingTop: '80px'}}><Cart /></div>} />

          {/* Admin Routes */}

          <Route path="/admin/dashboard" element={<AdminDashboard />} />

          <Route path="/admin/users" element={<AdminUsers />} />

          <Route path="/admin/instructors" element={<AdminInstructors />} />

          <Route path="/admin/earnings" element={<AdminEarnings />} />

          <Route path="/admin/profile" element={<AdminProfile />} />

          <Route path="/admin/settings" element={<AdminSettings />} />

          {/* Instructor Routes */}

          <Route path="/instructor/dashboard" element={<InstructorDashboard />} />

          <Route path="/instructor/courses" element={<InstructorCourses />} />

          <Route path="/instructor/courses/:id" element={<InstructorCourseDetails />} />

          <Route path="/instructor/students" element={<InstructorStudents />} />

          <Route path="/instructor/profile" element={<InstructorProfile />} />

          <Route path="/instructor/settings" element={<InstructorSettings />} />

        </Routes>

      </main>

      <FooterWrapper />

      <ChatbotWrapper />

    </div>

  </Router>);}

export default App;
```

Server.js

```
const express = require('express');

const mongoose = require('mongoose');

const cors = require('cors');

const dotenv = require('dotenv');

dotenv.config();

const app = express();

const PORT = process.env.PORT || 5000;

// Middleware

app.use(express.json());

app.use(cors());

// Database Connection

const MONGO_URI = process.env.MONGO_URI || 'mongodb://localhost:27017/learning-platform';

mongoose.connect(MONGO_URI)

    .then(() => console.log('MongoDB Connected'))

    .catch(err => console.error('MongoDB connection error:', err));

// Routes

app.use('/api/auth', require('./routes/auth'));

app.use('/api/courses', require('./routes/courses'));

app.use('/api/admin', require('./routes/admin'));

app.use('/api/users', require('./routes/users'));

app.use('/api/contact', require('./routes/contact'));

app.use('/api/instructors', require('./routes/instructors'));

app.use('/api/cart', require('./routes/cart'));

app.use('/api/notifications', require('./routes/notifications'));

app.use('/api/chatbot', require('./routes/chatbot'));
```

```
app.get('/', (req, res) => {
   res.send('API is running...');
});
app.listen(PORT, () => {
   console.log(`Server running on port ${PORT}`);
});
```

Cloudinary.js

```
const cloudinary = require('cloudinary').v2;
const { CloudinaryStorage } = require('multer-storage-cloudinary');
const multer = require('multer');
const dotenv = require('dotenv');
dotenv.config();
cloudinary.config({
  cloud_name: process.env.CLOUDINARY_CLOUD_NAME,
  api_key: process.env.CLOUDINARY_API_KEY,
  api_secret: process.env.CLOUDINARY_API_SECRET,
});
const storage = new CloudinaryStorage({
  cloudinary: cloudinary,
  params: {
    folder: 'resumes', // The folder in Cloudinary
    allowed_formats: ['pdf'], // Restrict to PDF
    resource_type: 'auto', // Let Cloudinary decide (usually 'image' for valid PDFs, 'raw' for others)
    format: async (req, file) => 'pdf',
```

```
    public_id: (req, file) => `resume_${req.params.userId || 'user'}_${Date.now()}`,

    access_mode: 'public', // Explicitly request public access},});

const courseStorage = new CloudinaryStorage({

    cloudinary: cloudinary,

    params: {

        folder: 'course_content',

        resource_type: 'auto', // Important for video/pdf

        allowed_formats: ['jpg', 'png', 'pdf', 'mp4', 'mkv', 'mov'],

        public_id: (req, file) => `course_${req.params.id ||
'content'}_${Date.now()}_${file.originalname.split('.')[0]}` }});

const upload = multer({ storage: storage });

const courseUpload = multer({ storage: courseStorage });

const submissionStorage = new CloudinaryStorage({

    cloudinary: cloudinary,

    params: {

        folder: 'course_submissions',

        resource_type: 'auto',

        allowed_formats: ['pdf', 'doc', 'docx', 'zip', 'jpg', 'png'],

        public_id: (req, file) =>
`submission_${req.params.userId}_${req.params.assignmentId}_${Date.now()}`}});

const submissionUpload = multer({ storage: submissionStorage });

module.exports = { upload, courseUpload, submissionUpload, cloudinary }
```

9.2 SCREENSHOTS

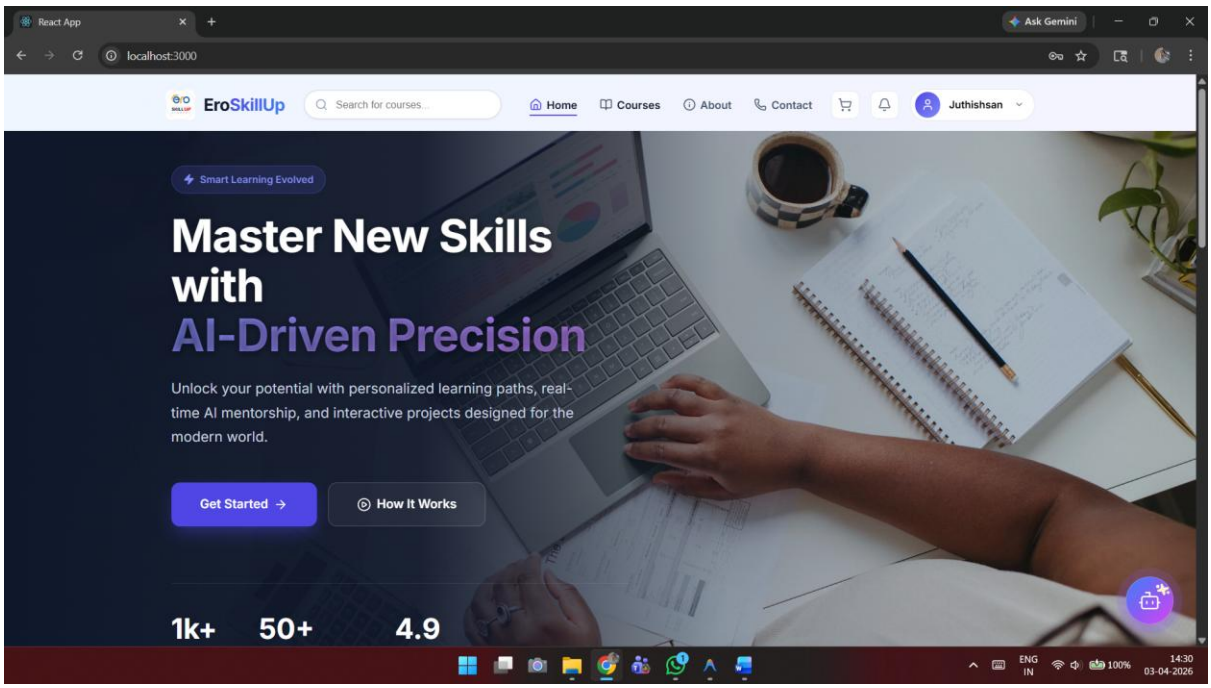

Figure 9.2.1 Learner Home Page

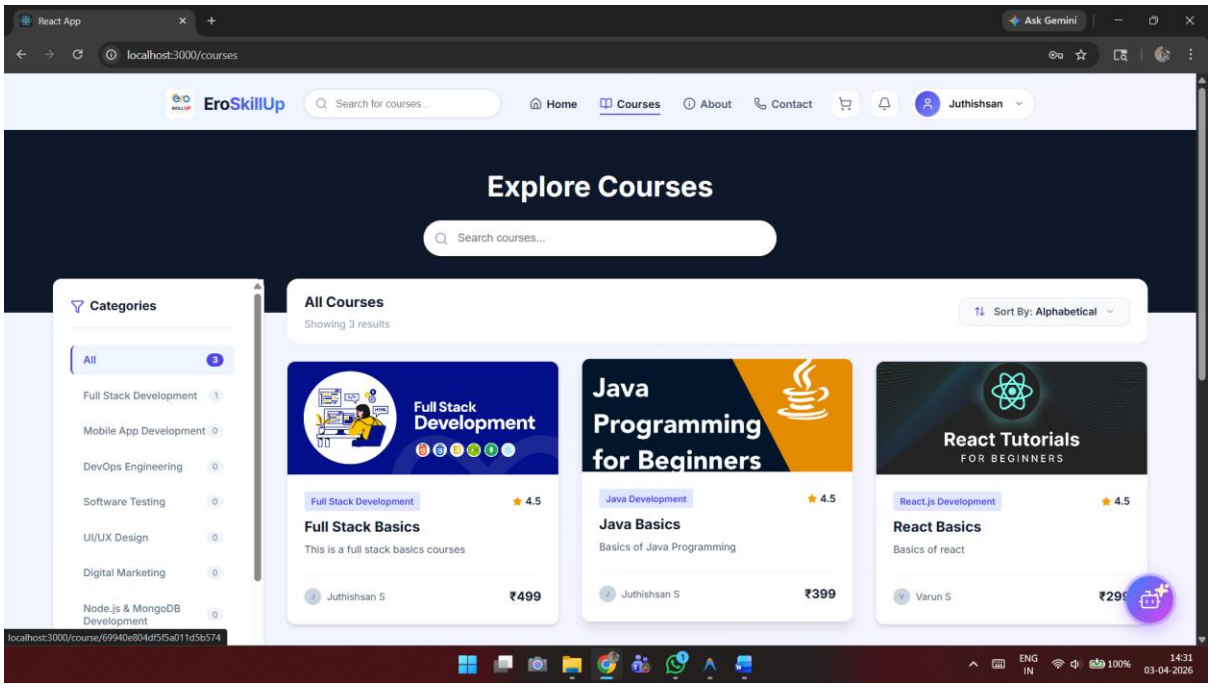

Figure 9.2.2 Course Page

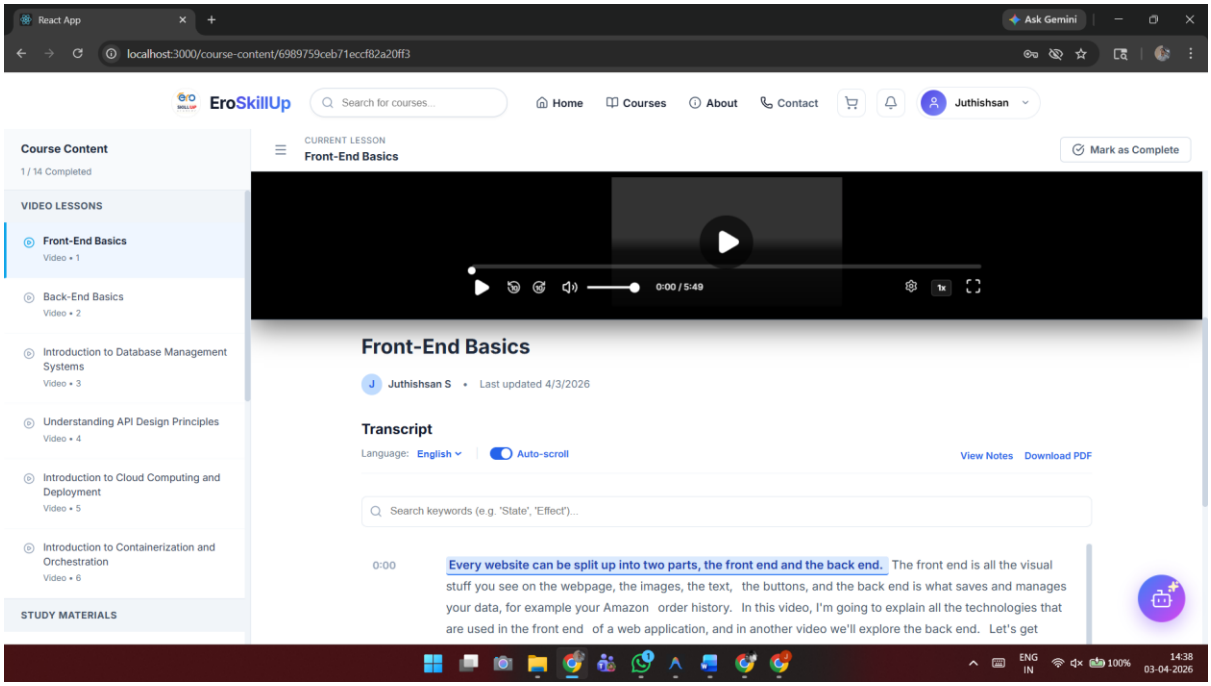

Figure 9.2.3 Course Content Page

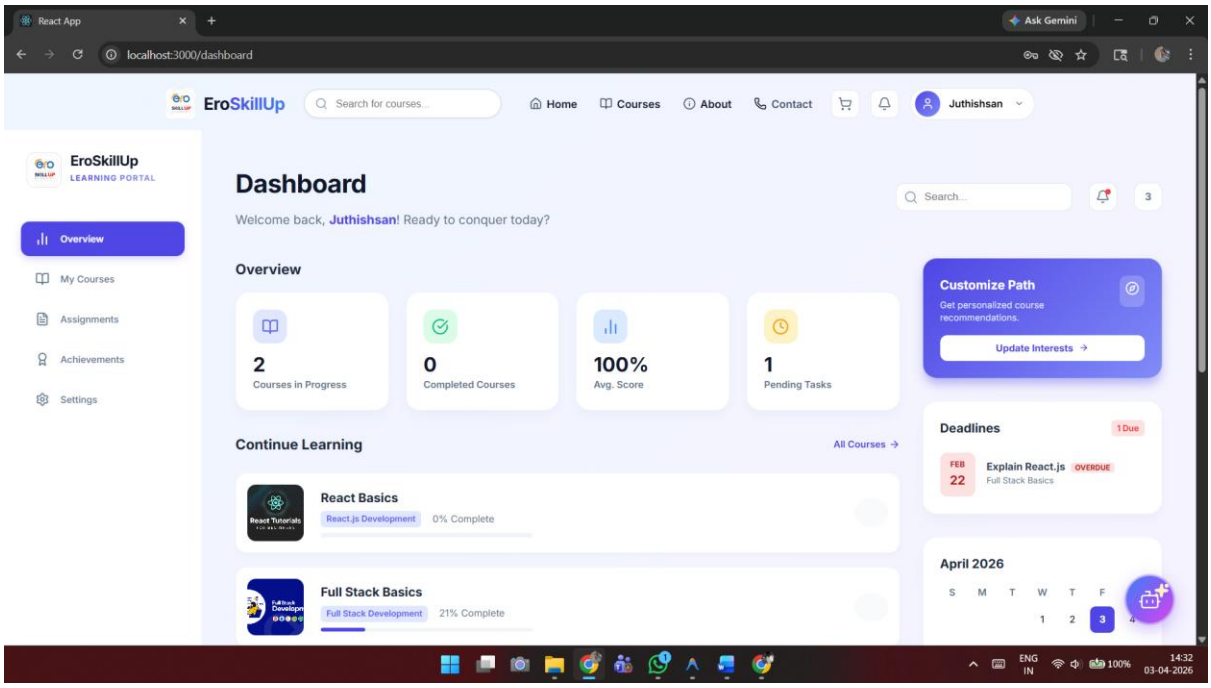

Figure 9.2.4 Learner Dashboard

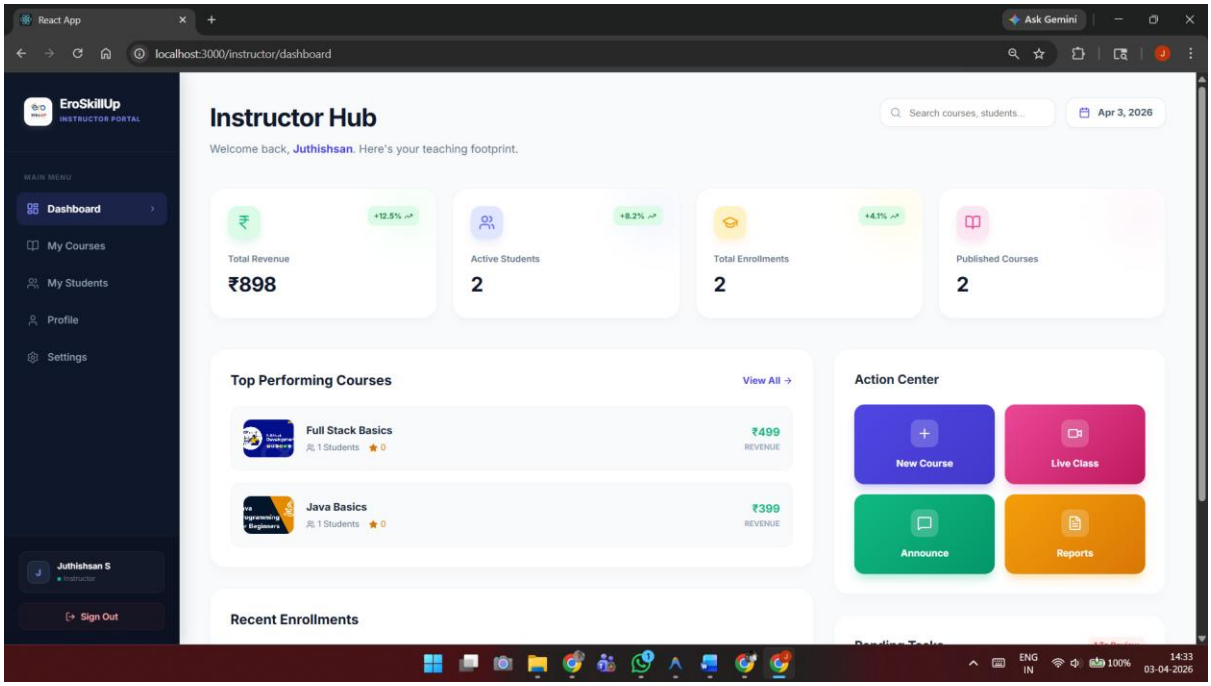

Figure 9.2.5 Instructor Dashboard

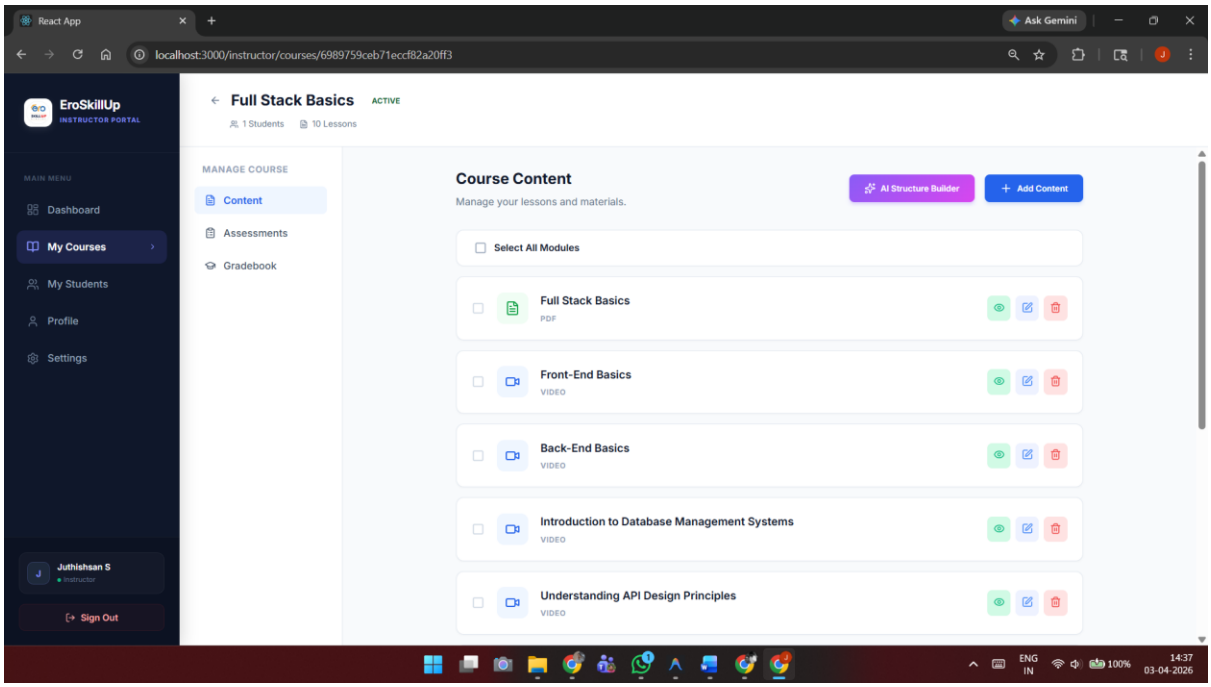

Figure 9.2.6 Instructor Course Management

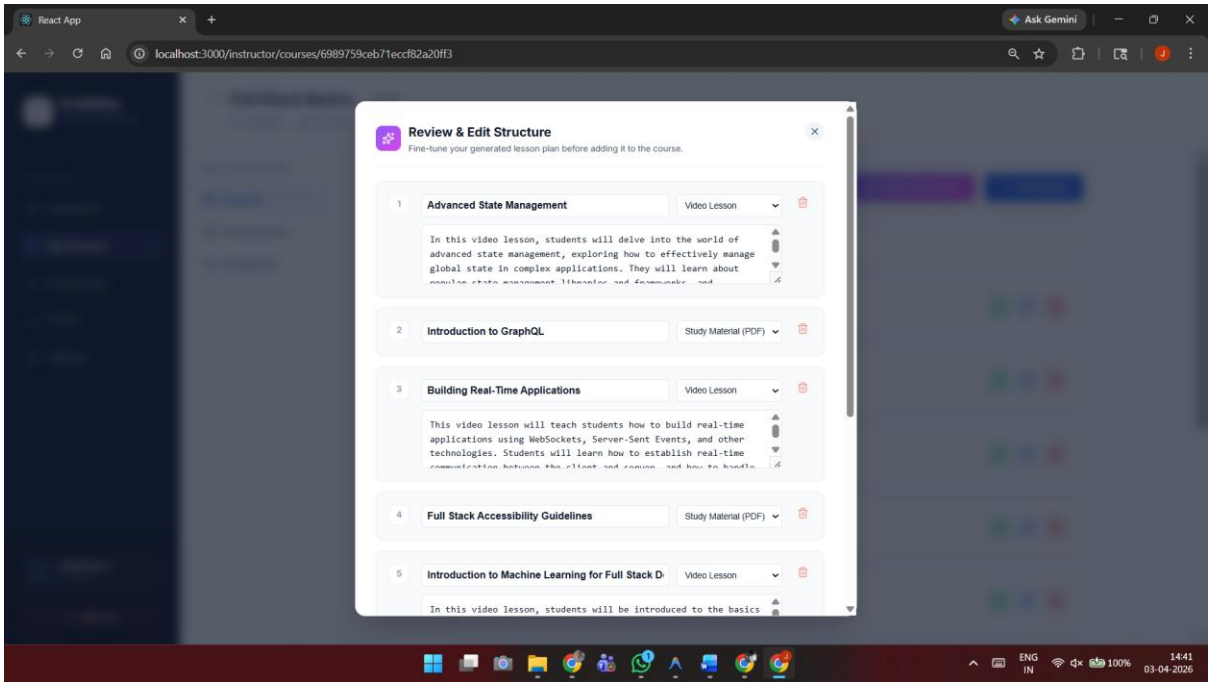

Figure 9.2.7 AI Course Content Builder

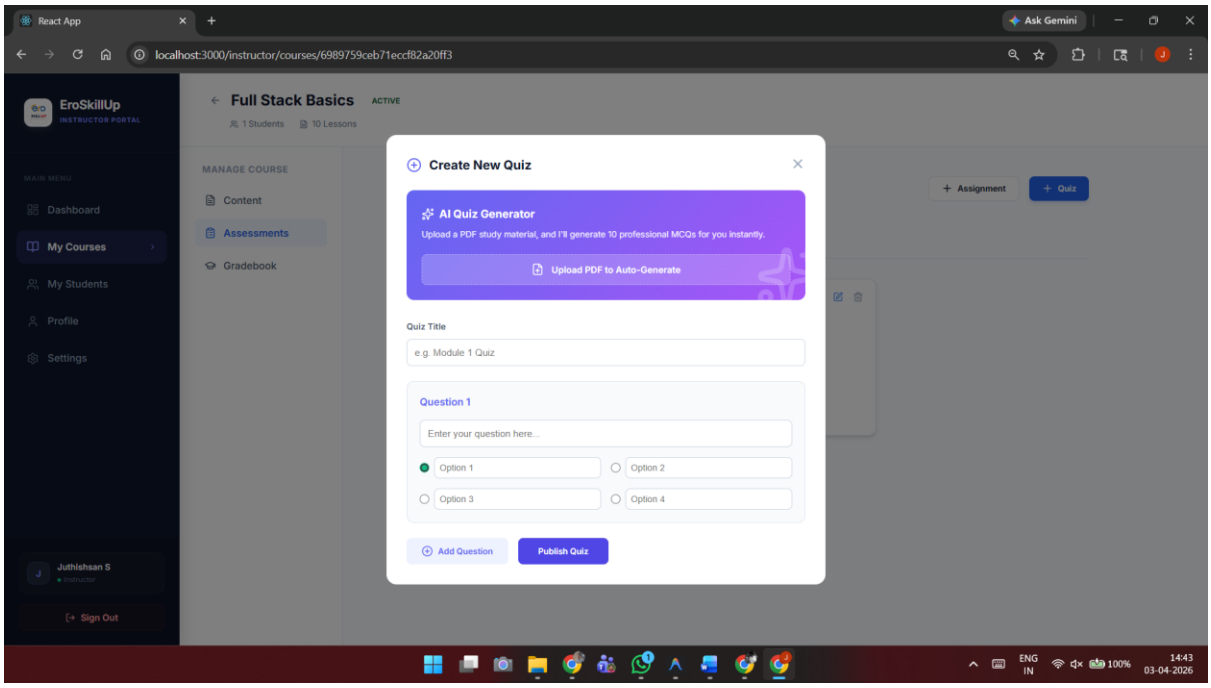

Figure 9.2.8 AI Quiz Generator

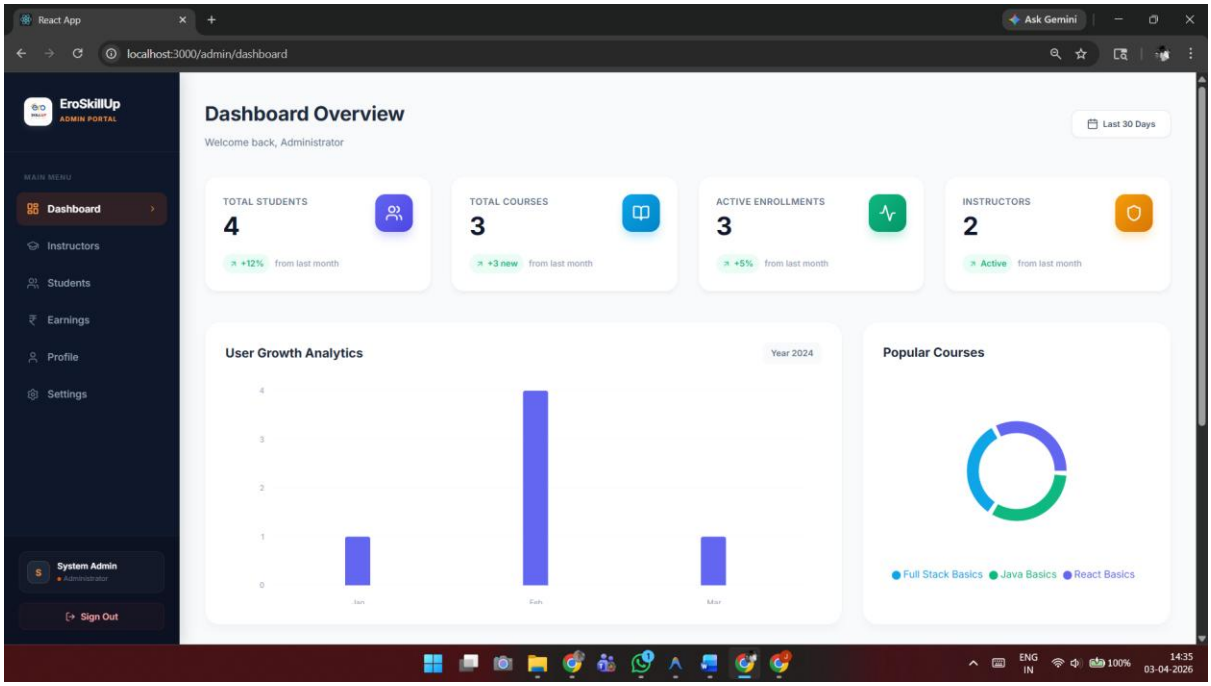

Figure 9.2.9 Admin Dashboard

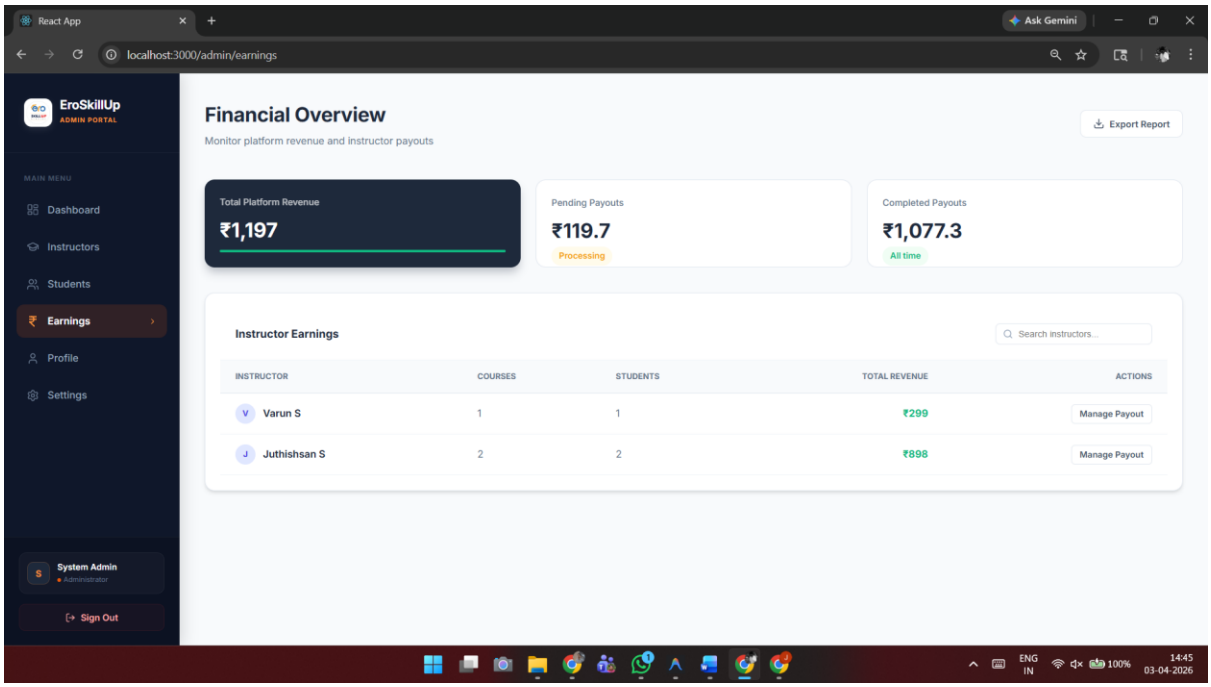

Figure 9.2.10 Financial Management

CHAPTER 10

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